Assignment-2

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Aim: DDL and DML queries here regarding
1. Case study: Airline management system and
2. Mini project (E-commerce)

1.AIRLINE MANAGEMENT SYSTEM

1. Creating Database

```
CREATE DATABASE AirlineDB;
```

2. Selecting Database

```
USE AirlineDB;
```

3. Creating Table: Passengers

```
CREATE TABLE Passengers (
    passenger_id INT PRIMARY KEY,
    name VARCHAR(50),
    age INT,
    gender VARCHAR(10)
);
```

4. Creating Table: Airlines

```
CREATE TABLE Airlines (
    airline_id INT PRIMARY KEY,
    airline_name VARCHAR(50)
```

```
);
```

5. Creating Table: Flights

```
CREATE TABLE Flights (
    flight_id INT PRIMARY KEY,
    airline_id INT,
    source VARCHAR(50),
    destination VARCHAR(50),
    departure_time DATETIME,
    arrival_time DATETIME,
    FOREIGN KEY (airline_id) REFERENCES Airlines(airline_id)
);
```

6. Creating Table: Bookings

```
CREATE TABLE Bookings (
    booking_id INT PRIMARY KEY,
    passenger_id INT,
    flight_id INT,
    booking_date DATE,
    seat_no VARCHAR(10),
    price DECIMAL(10,2),
    FOREIGN KEY (passenger_id) REFERENCES Passengers(passenger_id),
    FOREIGN KEY (flight_id) REFERENCES Flights(flight_id)
);
```

7. Creating Table: Payments

```
CREATE TABLE Payments (
    payment_id INT PRIMARY KEY,
    booking_id INT,
    amount DECIMAL(10,2),
    payment_mode VARCHAR(20),
    FOREIGN KEY (booking_id) REFERENCES Bookings(booking_id)
);
```

DML (INSERT DATA)

8. Inserting Data into Passengers

```
INSERT INTO Passengers VALUES
(1,'Soham',21,'Male'),
(2,'Amit',25,'Male'),
(3,'Neha',23,'Female'),
(4,'Riya',22,'Female'),
(5,'Karan',24,'Male'),
(6,'Pooja',26,'Female');
```

9. Inserting Data into Airlines

```
INSERT INTO Airlines VALUES
(1,'IndiGo'),
(2,'Air India'),
(3,'SpiceJet');
```

10. Inserting Data into Flights

```
INSERT INTO Flights VALUES
(101,1,'Pune','Delhi','2026-04-01 10:00:00','2026-04-01 12:30:00'),
(102,2,'Mumbai','Bangalore','2026-04-02 09:00:00','2026-04-02 11:00:00'),
(103,1,'Delhi','Kolkata','2026-04-03 14:00:00','2026-04-03 16:30:00'),
(104,3,'Pune','Delhi','2026-04-04 08:00:00','2026-04-04 10:30:00'),
(105,2,'Mumbai','Bangalore','2026-04-05 15:00:00','2026-04-05 17:00:00'),
(106,3,'Pune','Chennai','2026-04-06 07:00:00','2026-04-06 09:30:00');
```

11. Inserting Data into Bookings

```
INSERT INTO Bookings VALUES
(201,1,101,'2026-03-25','A1',5000),
```

```
(202,2,102,'2026-03-26','B2',4500),
(203,3,101,'2026-03-27','A2',5000),
(204,4,103,'2026-03-28','C3',6000),
(205,5,104,'2026-03-29','D4',5200),
(206,6,105,'2026-03-30','E5',4800),
(207,2,101,'2026-03-30','A3',5000),
(208,3,104,'2026-03-31','D5',5200),
(209,1,106,'2026-04-01','F6',5500);
```

12. Inserting Data into Payments

```
INSERT INTO Payments VALUES
(301,201,5000,'UPI'),
(302,202,4500,'Card'),
(303,203,5000,'Cash'),
(304,204,6000,'UPI'),
(305,205,5200,'Card'),
(306,206,4800,'UPI'),
(307,207,5000,'Cash'),
(308,208,5200,'UPI'),
(309,209,5500,'Card');
```

Basic DML Queries

13. Display All Passengers

```
SELECT * FROM Passengers;
```

14. Display All Flights

```
SELECT * FROM Flights;
```

15. Select Specific Column

```
SELECT name, age FROM Passengers;
```

16. WHERE Condition (Price > 5000)

```
SELECT * FROM Bookings
WHERE price > 5000;
```

17. ORDER BY (Ascending)

```
SELECT * FROM Flights
ORDER BY source ASC;
```

18. ORDER BY (Descending)

```
SELECT * FROM Flights
ORDER BY destination DESC;
```

19. GROUP BY (Bookings per Flight)

```
SELECT flight_id, COUNT(*)
FROM Bookings
GROUP BY flight_id;
```

20. Aggregate Function (AVG Price)

```
SELECT AVG(price) FROM Bookings;
```

21. UPDATE Query

```
UPDATE Bookings
SET price = 5500
WHERE booking_id = 201;
```

22. DELETE Query

```
DELETE FROM Payments
```

WHERE payment_id = 302;

2. E-commerce(Shopsphere)

1. Creating Database

CREATE DATABASE shopsphere;

2. Selecting Database

USE shopsphere;

3. Creating Table: users

```
mysql> CREATE TABLE users (
    ->     id INT AUTO_INCREMENT PRIMARY KEY,
    ->     name VARCHAR(100),
    ->     email VARCHAR(100) UNIQUE,
    ->     password VARCHAR(100),
    ->     role VARCHAR(20),
    ->     phone VARCHAR(15),
    ->     avatar_url VARCHAR(255),
    ->     reset_token VARCHAR(255),
    ->     reset_token_expires DATETIME,
    ->     created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
    -> );
Query OK, 0 rows affected (0.09 sec)
```

4. Creating Table: categories

```
mysql> CREATE TABLE categories (
    ->     id INT AUTO_INCREMENT PRIMARY KEY,
    ->     name VARCHAR(100) UNIQUE,
    ->     slug VARCHAR(120) UNIQUE
    -> );
Query OK, 0 rows affected (0.05 sec)
```

5. Creating Table: products

```
mysql> CREATE TABLE products (
    ->     id INT AUTO_INCREMENT PRIMARY KEY,
    ->     category_id INT,
    ->     name VARCHAR(160),
    ->     slug VARCHAR(180) UNIQUE,
    ->     brand VARCHAR(120),
    ->     description TEXT,
    ->     price DECIMAL(10,2),
    ->     discount_percent INT DEFAULT 0,
    ->     stock INT DEFAULT 0,
    ->     image_url VARCHAR(255),
    ->     rating DECIMAL(3,2) DEFAULT 0.00,
    ->     review_count INT DEFAULT 0,
    ->     featured TINYINT(1) DEFAULT 0,
    ->     created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    ->     FOREIGN KEY (category_id) REFERENCES categories(id)
    -> );
```

6. Creating Table: orders

```
mysql> CREATE TABLE orders (
    ->     id INT AUTO_INCREMENT PRIMARY KEY,
    ->     user_id INT,
    ->     total_amount DECIMAL(10,2),
    ->     payment_method VARCHAR(60),
    ->     payment_status ENUM('pending','paid','failed') DEFAULT 'pending',
    ->     order_status ENUM('Pending','Shipped','Delivered') DEFAULT 'Pending',
    ->     shipping_address JSON,
    ->     created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    ->     FOREIGN KEY (user_id) REFERENCES users(id)
    -> );
Query OK, 0 rows affected (0.06 sec)
```

7. Creating Table: order_items

```
mysql> CREATE TABLE order_items (
    ->     id INT AUTO_INCREMENT PRIMARY KEY,
    ->     order_id INT,
    ->     product_id INT,
    ->     quantity INT,
    ->     price DECIMAL(10,2),
    ->     FOREIGN KEY (order_id) REFERENCES orders(id),
    ->     FOREIGN KEY (product_id) REFERENCES products(id)
    -> );
Query OK, 0 rows affected (0.07 sec)
```

8. Creating Table: cart_items

```
mysql> CREATE TABLE cart_items (
    ->     id INT AUTO_INCREMENT PRIMARY KEY,
    ->     user_id INT,
    ->     product_id INT,
    ->     quantity INT DEFAULT 1,
    ->     FOREIGN KEY (user_id) REFERENCES users(id),
    ->     FOREIGN KEY (product_id) REFERENCES products(id)
    -> );
Query OK, 0 rows affected (0.07 sec)
```

9. Creating Table: reviews

```
mysql> CREATE TABLE reviews (
    ->     id INT AUTO_INCREMENT PRIMARY KEY,
    ->     user_id INT,
    ->     product_id INT,
    ->     rating INT,
    ->     comment TEXT,
    ->     created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    ->     FOREIGN KEY (user_id) REFERENCES users(id),
    ->     FOREIGN KEY (product_id) REFERENCES products(id)
    -> );
Query OK, 0 rows affected (0.07 sec)
```

10. Inserting Data into categories

```
mysql> INSERT INTO categories (name,slug) VALUES
    -> ('Electronics','electronics'),
    -> ('Clothing','clothing'),
    -> ('Books','books');
Query OK, 3 rows affected (0.01 sec)
```

11. Inserting Data into products

```
mysql> INSERT INTO products (category_id,name,slug,brand,price,stock)
VALUES
    -> (1,'Smartphone','smartphone','Samsung',20000,10),
    -> (1,'Laptop','laptop','Dell',50000,5),
    -> (2,'T-Shirt','tshirt','Nike',1000,20),
    -> (3,'Database Book','db-book','Pearson',500,15);
Query OK, 4 rows affected (0.01 sec)
```

12. Inserting Data into users

```
mysql> INSERT INTO users (id,name,email,password,role,phone)
    -> VALUES
    -> (1,'Soham','soham@gmail.com','pass123','customer','9876543210'),
    -> (2,'Amit','amit@gmail.com','pass123','customer','9876543211'),
    -> (3,'Neha','neha@gmail.com','pass123','customer','9876543212');
Query OK, 3 rows affected (0.01 sec)
```

13. Inserting Data into orders

```
mysql> INSERT INTO orders
(id,user_id,total_amount,payment_method,payment_status,order_status,shipping
_address)
    -> VALUES
    -> (1,1,21000,'UPI','paid','Delivered','{"city":"Pune"}'),
    -> (2,2,50000,'Card','paid','Shipped','{"city":"Mumbai"}');
Query OK, 2 rows affected (0.01 sec)
```

14. Inserting Data into order_items

```
mysql> INSERT INTO order_items (order_id,product_id,quantity,price) VALUES
    -> (1,1,1,20000),
    -> (1,3,1,1000),
    -> (2,2,1,50000);
Query OK, 3 rows affected (0.01 sec)
```

15. Inserting Data into cart_items

```
mysql> INSERT INTO cart_items (user_id,product_id,quantity) VALUES
    -> (1,2,1),
    -> (2,1,2),
    -> (3,4,1);
Query OK, 3 rows affected (0.01 sec)
```

16. Inserting Data into reviews

```
mysql> INSERT INTO reviews (user_id,product_id,rating,comment) VALUES
    -> (1,1,5,'Excellent product'),
```

```
    -> (2,2,4,'Very good laptop'),
    -> (3,4,5,'Great book for DBMS');
Query OK, 3 rows affected (0.01 sec)
```

17. Displaying Tables (SHOW TABLES)

```
mysql> SHOW TABLES;
+-----------------------+
| Tables_in_shopsphere |
+-----------------------+
| cart_items            |
| categories            |
| order_items           |
| orders                |
| products              |
| reviews               |
| users                 |
+-----------------------+
7 rows in set (0.02 sec)
```

18. Select All Records from users

```
mysql> SELECT * FROM users;
+----+-------+-----------------+----------+----------+------------+------------+-------------+---------------------+---------------------+
| id | name  | email           | password | role     | phone      | avatar_url | reset_token | reset_token_expires | created_at          |
+----+-------+-----------------+----------+----------+------------+------------+-------------+---------------------+---------------------+
|  1 | Soham | soham@gmail.com | pass123  | customer | 9876543210 | NULL       | NULL        | NULL                | 2026-04-13 20:06:55 |
|  2 | Amit  | amit@gmail.com  | pass123  | customer | 9876543211 | NULL       | NULL        | NULL                | 2026-04-13 20:06:55 |
|  3 | Neha  | neha@gmail.com  | pass123  | customer | 9876543212 | NULL       | NULL        | NULL                | 2026-04-13 20:06:55 |
+----+-------+-----------------+----------+----------+------------+------------+-------------+---------------------+---------------------+
3 rows in set (0.00 sec)
```

19. Select All Records from products

```
mysql> SELECT * FROM products;
+----+-------------+---------------+------------+---------+-------------+----------+-----------------+-------+-----------+--------+--------------+----------+---------
-----------+
| id | category_id | name          | slug       | brand   | description | price    | discount_percent | stock | image_url | rating | review_count | featured | created_a
t          |
+----+-------------+---------------+------------+---------+-------------+----------+-----------------+-------+-----------+--------+--------------+----------+---------
-----------+
|  1 |           1 | Smartphone    | smartphone | Samsung | NULL        | 20000.00 |               0 |    10 | NULL      |   0.00 |            0 |        0 | 2026-04-1
3 20:04:12 |
|  2 |           1 | Laptop        | laptop     | Dell    | NULL        | 50000.00 |               0 |     5 | NULL      |   0.00 |            0 |        0 | 2026-04-1
3 20:04:12 |
|  3 |           2 | T-Shirt       | tshirt     | Nike    | NULL        |  1000.00 |               0 |    20 | NULL      |   0.00 |            0 |        0 | 2026-04-1
3 20:04:12 |
|  4 |           3 | Database Book | db-book    | Pearson | NULL        |   500.00 |               0 |    15 | NULL      |   0.00 |            0 |        0 | 2026-04-1
3 20:04:12 |
+----+-------------+---------------+------------+---------+-------------+----------+-----------------+-------+-----------+--------+--------------+----------+---------
-----------+
4 rows in set (0.00 sec)
```

20. Select Specific Columns (name, price)

```
mysql> SELECT name, price FROM products;
+----------------+----------+
| name           | price    |
+----------------+----------+
| Smartphone     | 20000.00 |
| Laptop         | 50000.00 |
| T-Shirt        |  1000.00 |
| Database Book  |   500.00 |
+----------------+----------+
4 rows in set (0.00 sec)
```

21. Select with WHERE Condition (price > 10000)

```
mysql> SELECT * FROM products
    -> WHERE price > 10000;
+----+-------------+------------+------------+---------+-------------+----------+-----------------+-------+-----------+--------+--------------+----------+-------------
--------+
| id | category_id | name       | slug       | brand   | description | price    | discount_percent | stock | image_url | rating | review_count | featured | created_at
        |
+----+-------------+------------+------------+---------+-------------+----------+-----------------+-------+-----------+--------+--------------+----------+-------------
--------+
|  1 |           1 | Smartphone | smartphone | Samsung | NULL        | 20000.00 |               0 |    10 | NULL      |   0.00 |            0 |        0 | 2026-04-13 2
0:04:12 |
|  2 |           1 | Laptop     | laptop     | Dell    | NULL        | 50000.00 |               0 |     5 | NULL      |   0.00 |            0 |        0 | 2026-04-13 2
0:04:12 |
+----+-------------+------------+------------+---------+-------------+----------+-----------------+-------+-----------+--------+--------------+----------+-------------
--------+
2 rows in set (0.00 sec)
```

22. Select with WHERE Condition (role = customer)

```
mysql> SELECT * FROM users
    -> WHERE role = 'customer';
+----+-------+-----------------+----------+----------+------------+------------+-------------+---------------------+---------------------+
| id | name  | email           | password | role     | phone      | avatar_url | reset_token | reset_token_expires | created_at          |
+----+-------+-----------------+----------+----------+------------+------------+-------------+---------------------+---------------------+
|  1 | Soham | soham@gmail.com | pass123  | customer | 9876543210 | NULL       | NULL        | NULL                | 2026-04-13 20:06:55 |
|  2 | Amit  | amit@gmail.com  | pass123  | customer | 9876543211 | NULL       | NULL        | NULL                | 2026-04-13 20:06:55 |
|  3 | Neha  | neha@gmail.com  | pass123  | customer | 9876543212 | NULL       | NULL        | NULL                | 2026-04-13 20:06:55 |
+----+-------+-----------------+----------+----------+------------+------------+-------------+---------------------+---------------------+
3 rows in set (0.00 sec)
```

23. Select Orders with payment_status = 'paid'

```
mysql> SELECT * FROM orders
    -> WHERE payment_status = 'paid';
+----+---------+--------------+----------------+----------------+--------------+---------------------+---------------------+
| id | user_id | total_amount | payment_method | payment_status | order_status | shipping_address    | created_at          |
+----+---------+--------------+----------------+----------------+--------------+---------------------+---------------------+
|  1 |       1 |     21000.00 | UPI            | paid           | Delivered    | {"city": "Pune"}    | 2026-04-13 20:07:41 |
|  2 |       2 |     50000.00 | Card           | paid           | Shipped      | {"city": "Mumbai"}  | 2026-04-13 20:07:41 |
+----+---------+--------------+----------------+----------------+--------------+---------------------+---------------------+
2 rows in set (0.00 sec)
```

24. Sorting Data using ORDER BY (Ascending)

```
mysql> SELECT * FROM products
    -> ORDER BY price ASC;
+----+-------------+---------------+------------+---------+-------------+----------+-----------------+-------+-----------+--------+--------------+----------+----------
-----------+
| id | category_id | name          | slug       | brand   | description | price    | discount_percent | stock | image_url | rating | review_count | featured | created_a
t          |
+----+-------------+---------------+------------+---------+-------------+----------+-----------------+-------+-----------+--------+--------------+----------+----------
----------+
|  4 |           3 | Database Book | db-book    | Pearson | NULL        |   500.00 |               0 |    15 | NULL      |   0.00 |            0 |        0 | 2026-04-1
3 20:04:12 |
|  3 |           2 | T-Shirt       | tshirt     | Nike    | NULL        |  1000.00 |               0 |    20 | NULL      |   0.00 |            0 |        0 | 2026-04-1
3 20:04:12 |
|  1 |           1 | Smartphone    | smartphone | Samsung | NULL        | 20000.00 |               0 |    10 | NULL      |   0.00 |            0 |        0 | 2026-04-1
3 20:04:12 |
|  2 |           1 | Laptop        | laptop     | Dell    | NULL        | 50000.00 |               0 |     5 | NULL      |   0.00 |            0 |        0 | 2026-04-1
3 20:04:12 |
+----+-------------+---------------+------------+---------+-------------+----------+-----------------+-------+-----------+--------+--------------+----------+----------
-----------+
4 rows in set (0.00 sec)
```

25. Sorting Data using ORDER BY (Descending)

```
mysql> SELECT * FROM products
    -> ORDER BY price DESC;
+----+-------------+---------------+------------+---------+-------------+----------+-----------------+-------+-----------+--------+--------------+----------+----------
----------+
| id | category_id | name          | slug       | brand   | description | price    | discount_percent | stock | image_url | rating | review_count | featured | created_a
t          |
+----+-------------+---------------+------------+---------+-------------+----------+-----------------+-------+-----------+--------+--------------+----------+----------
----------+
|  2 |           1 | Laptop        | laptop     | Dell    | NULL        | 50000.00 |               0 |     5 | NULL      |   0.00 |            0 |        0 | 2026-04-1
3 20:04:12 |
|  1 |           1 | Smartphone    | smartphone | Samsung | NULL        | 20000.00 |               0 |    10 | NULL      |   0.00 |            0 |        0 | 2026-04-1
3 20:04:12 |
|  3 |           2 | T-Shirt       | tshirt     | Nike    | NULL        |  1000.00 |               0 |    20 | NULL      |   0.00 |            0 |        0 | 2026-04-1
3 20:04:12 |
|  4 |           3 | Database Book | db-book    | Pearson | NULL        |   500.00 |               0 |    15 | NULL      |   0.00 |            0 |        0 | 2026-04-1
3 20:04:12 |
+----+-------------+---------------+------------+---------+-------------+----------+-----------------+-------+-----------+--------+--------------+----------+----------
----------+
4 rows in set (0.00 sec)
```

26. Updating Records in products

```
mysql> UPDATE products
    -> SET stock = 50
    -> WHERE id = 1;
Query OK, 1 row affected (0.01 sec)
```

27. Deleting Record from cart_items

```
mysql> DELETE FROM cart_items
    -> WHERE id = 2;
Query OK, 1 row affected (0.01 sec)
```

28. GROUP BY Query (Count products per category)

```
mysql> SELECT category_id, COUNT(*)
    -> FROM products
    -> GROUP BY category_id;
+-------------+----------+
| category_id | COUNT(*) |
+-------------+----------+
|           1 |        2 |
|           2 |        1 |
|           3 |        1 |
+-------------+----------+
3 rows in set (0.00 sec)
```

29. Aggregate Function: AVG(price)

```
mysql> SELECT AVG(price) FROM products;
+--------------+
| AVG(price)   |
+--------------+
| 17875.000000 |
+--------------+
1 row in set (0.00 sec)
```

30. Aggregate Function: MAX(price)

```
mysql> SELECT MAX(price) FROM products;
+------------+
| MAX(price) |
+------------+
|   50000.00 |
+------------+
1 row in set (0.00 sec)
```

31. Aggregate Function: MIN(price)

```
mysql> SELECT MIN(price) FROM products;
+------------+
| MIN(price) |
+------------+
|     500.00 |
+------------+
1 row in set (0.00 sec)
```